

Amit Kumar Mohanty

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 1+ year of experience architecting production-grade AI systems, high-throughput model-serving platforms, and event-driven applications. Proven track record of leading technical initiatives, designing low-latency services at enterprise scale, and delivering measurable business impact through ML engineering and AI-driven automation. Outside of work, a competitive tennis player and avid sketch artist, pursuits that sharpen discipline, focus, and creativity.

EDUCATION

Birla Institute of Technology and Science, Pilani (BITS Pilani) July 2026 – July 2028
M.Tech in Artificial Intelligence and Machine Learning (Currently Pursuing)
Relevant Coursework: NLP, Deep Learning, Generative AI, LLMs, Deep Neural Networks, Deep Reinforcement Learning
K.R. Mangalam University July 2021 – July 2025
B.Tech in Computer Science and Engineering (Specialization in AI/ML), Gurgaon

EXPERIENCE

Research Associate – AI, Keywords Studios India – Gurgaon, Haryana May 2025 – Present

- Awarded “**Rookie of the Year 2025**” for surpassing **all performance KPIs** through advanced model training, data-pipeline optimization, and cross-functional collaboration.
- Trained and optimized **enterprise-grade Agentic AI** models to autonomously execute complex multi-step workflows (flight booking, e-commerce ordering, web data retrieval), improving **task-completion reliability**.
- Contributed to a cutting-edge **web-automation AI agent** that visually interprets on-screen content and, from a natural-language instruction, autonomously plans and executes browser actions such as clicking, typing, and navigating to complete complex, multi-step tasks.
- Trained, evaluated, and quality-controlled the agent with a **global team**, curating high-quality datasets and running structured test reviews to deliver client-ready iterations with **improved action accuracy and reliability**.
- Built custom data pipelines and delivered **85%** of self-generated training datasets directly to production as the **first member of a global team**, accelerating model readiness.

Artificial Intelligence Intern, Infosys – Remote November 2024 – February 2025

- Engineered an AI call-center system with **LangChain and LangSmith**, boosting model response accuracy by **35%** and elevating customer-AI interaction quality.
- Developed and deployed LLM features that reduced system latency by **25%**, streamlining the model deployment process.
- Designed and A/B-tested prompt-engineering strategies with automated LangSmith evaluation, reducing hallucinations by **20%** and making customer conversations feel more natural and trustworthy.
- Collaborated with a team of **12 engineers** to integrate end-to-end ML workflows and earned selection to present at the **Best Project by Indian Interns** initiative.

PROJECTS

Dynamic Screen Companion June 2025 – Present

- Built a real-time multi-modal assistant integrating the **Gemini API** to analyze live screen activity and deliver continuous voice feedback and interactive support.
- Engineered a high-performance FastAPI model-serving backend handling concurrent inference streams, reducing manual task time by **40%**.
- Designed the end-to-end multi-modal architecture from scratch and optimized OCR pipelines to maintain **95%+ extraction accuracy** across varied screen layouts.

Diabetic Optiscan January 2025 – June 2025

- Built an AI tool that scans retinal (eye) images to detect early signs of diabetic retinopathy, and co-authored a **peer-reviewed research paper** explaining how the system was designed and the accuracy improvements it achieved.
- Led integration of a **Vision Transformer (ViT-B/16)** model on the APTOS dataset, optimizing self-attention to capture complex spatial patterns in high-resolution retinal images.
- Designed a custom wavelet modification technique that boosted detection precision by **50%** over legacy methods.

Currency Detection App for Visually Impaired July 2024 – December 2024

- Trained an image-classification model on **200,000+** banknote images achieving **96% accuracy**, deployed in a mobile app with voice feedback to help prevent fraud.
- Conducted user interviews with visually impaired individuals to refine real-world accessibility features.

SKILLS

Languages & Frameworks: Python, C++, SQL, R, FastAPI, TensorFlow, PyTorch, OpenCV, Scikit-learn, LangChain, LangGraph
AI & Machine Learning: LLMs, Generative AI, Agentic AI, RAG, Fine-tuning (LoRA, QLoRA), Deep Learning, Transformers, Hugging Face, BERT, NLP, Prompt Engineering, Model Evaluation & Optimization, OOP, DSA
MLOps & Deployment: Docker, Kubernetes, CI/CD, MLflow, Model Serving, Model Monitoring, Data Pipelines, Git, Vector Databases (FAISS/Chroma)
Data & Cloud Tools: AWS, GCP (Vertex AI), Power BI, MySQL, MongoDB, Pandas, NumPy, Statistical Analysis

CERTIFICATIONS

Ducat India – Data Science Professional Training April 2024 – April 2025
Professional training in Data Science: Python, statistics, machine learning, data analysis, and predictive modeling.
AI/ML for Geodata Analysis – ISRO
Applied ML and deep learning to satellite and remote-sensing imagery for terrain classification and geospatial pattern analysis.